

To modify a zone, select the one in the *Zone* menu on the left. Network interfaces and source IP addresses/ranges can be assigned in the *Interfaces* and *Sources* tabs, respectively, which are on the right.

Ports can be opened by either putting a checkmark in front of them in the *Services* tab, or by adding a new port in the *Ports* tab for that zone.

If a specific set of ports has to be opened in multiple zones, a sysadmin can also define a service for those ports. This can be done in the *Services* tab at the top of the window.

The default zone for otherwise unspecified connections can be changed by going to *Options>Change>Default Zone*.

Important: Any changes made in the permanent configuration will not become active until the next time the firewalld service unit is restarted or reloaded. Likewise, any changes made in the runtime configurations will not survive a reload or restart of the firewalld service.

Configuring firewall settings with firewall-cmd

For those sysadmins who prefer to work on the command line or who cannot use a graphical environment for some reason, there is also a command line client to interact with the firewall, which is `firewall-cmd`.

Firewall-cmd is installed as part of the main firewalld package. Firewall-cmd can perform the same actions that `firewall-config` can.

Shown below are a number of frequently used `firewall-cmd` commands, along with explanations about them. Note that unless otherwise specified, almost all commands will work on the runtime configuration, unless the `--permanent` option is specified. Many of the commands listed take the `--zone=<ZONE>` option to determine which zone they affect.

- `--get-default-zone`: Queries the current default zone.
- `--set-default-zone=<ZONE>`: Sets the default zone. This changes both the runtime and the permanent configuration.
- `--get-zones`: Lists all available zones.
- `--get-active-zones`: Lists all zones currently in use (that have an interface or source tied to them), along with their interface and source information.
- `--add-source=<CIDR> [--zone=<ZONE>]`: Routes all traffic coming from the IP address or network/netmask `<CIDR>` to the specified zone. If no `--zone=` option is provided, the default zone will be used.
- `--remove-source=<CIDR> [--zone=<ZONE>]`: Removes the rule routing all traffic coming from the IP address or network/netmask `<CIDR>` from the specified zone. If no `--zone=` option is provided, the default zone will be used.
- `--add-interface=<INTERFACE> [--zone=<ZONE>]`: Routes all traffic coming from `<INTERFACE>` to the specified zone. If no `--zone=` option is provided, the default zone will be used.
- `--change-interface=<INTERFACE> [--zone=<ZONE>]`: Associates the interface with `<ZONE>` instead of its

current zone. If no `--zone=` option is provided, the default zone will be used.

- `--list-all [--zone=<ZONE>]`: Lists all configured interfaces, sources, services and ports for `<ZONE>`. If no `--zone=` option is provided, the default zone will be used.
- `--list-all-zones`: Retrieves all information for all zones (interfaces, sources, ports, services, etc).
- `--add-service=<SERVICE> [--zone=<ZONE>]`: Allows traffic to `<SERVICE>`. If no `--zone=` option is provided, the default zone will be used.
- `--add-port=<PORT/PROTOCOL> [--zone=<ZONE>]`: Allows traffic to the `<PORT/PROTOCOL>` ports. If no `--zone=` option is provided, the default zone will be used.
- `--remove-service=<SERVICE> [--zone=<ZONE>]`: Removes `<SERVICE>` from the allowed list for the zone. If no `--zone=` option is provided, the default zone will be used.
- `--remove-port=<PORT/PROTOCOL> [--zone=<ZONE>]`: Removes the `<PORT/PROTOCOL>` ports from the allowed list for the zone. If no `--zone=` option is provided, the default zone will be used.
- `--reload`: Drops the runtime configuration and applies the persistent change.

Firewall-cmd example

The following example shows the default zone being set to `dmz` — all traffic coming from the `192.168.0.0/24` network being associated to the internal zone, and the network ports for `mysql` being opened on the internal zone.

```
# firewall-cmd --set-default-zone=dmz
# firewall-cmd --permanent --zone=internal --add-source=192.168.0.0/24
# firewall-cmd --permanent --zone=internal --add-service=mysql
# firewall-cmd --reload
```



Note: For situations where the basic syntax of firewalld is not enough, systems administrators can also add rich rules and a more expressive syntax, to write more complex rules. If even the rich rule syntax is not enough, sysadmins can also use direct configuration rules, which is basically raw iptables syntax that will be mixed with the firewalld rules.

END

References

`firewall-cmd` (1), `firewall-config` (1), `firewalld` (1), `firewalld.zone` (5) and `firewalld.zones` (5) manual pages, which can be accessed by using the `man` command

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